

Information Theory

Part 1: How do we represent information and uncertainty mathematically?

Probabilistic Machine Learning

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From predicting tokens to measuring uncertainty

We built models that predict the next token in a sequence.

Two questions

- How predictable is a piece of text?
- How do we score a model's predictions against the truth?

In machine learning, information theory helps us understand relationships between knowns and unknowns, design loss functions, and establish limits on what any algorithm can do with a given amount of data.

Both questions are answered in bits

Predictability and prediction quality both reduce to counting bits.

Outline

- 1 Quantifying uncertainty: Entropy
- 2 Entropy in action: data compression
- 3 The wrong distribution: relative entropy
- 4 More than one variable: Joint entropy and mutual information
- 5 Can processing add relevant information?
- 6 Limits on prediction

Rain in Seattle, rain in Phoenix

Let X = rain in Seattle, Y = rain in Phoenix (Bernoulli variables).

Question: How much information does each of these outcomes carry?

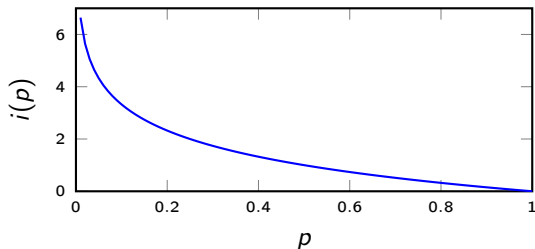
- Rain in Seattle ($X = 1$): near 50/50.
- No rain in Phoenix ($Y = 0$): high probability.
- Rain in Phoenix ($Y = 1$): low probability.

Rain in Seattle, rain in Phoenix

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Question: How much information does each of these outcomes carry?

- Rain in Seattle ($X = 1$): near 50/50. A fair amount of information.
- No rain in Phoenix ($Y = 0$): high probability. Little information.
- Rain in Phoenix ($Y = 1$): low probability. A lot of information.



Information decreases as probability increases

Self-information $i(p) = \log \frac{1}{p}$ (base 2, in bits).

Entropy

The information content of the statement ' $X = x_i$ ' is $i(p(x_i)) = \log \frac{1}{p(x_i)}$. Averaging over all outcomes gives the **entropy** of X :

$$H(X) = \mathbb{E} \left[\log \frac{1}{p(X)} \right] = \sum_{i=1}^m p(x_i) \log \frac{1}{p(x_i)}.$$

Range

$$0 \leq H(X) \leq \log m,$$

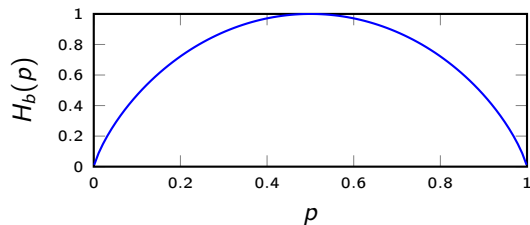
zero when the outcome is certain, $\log m$ when all m outcomes are equally likely.

Example: a fair die has $H = \log_2 6 \simeq 2.58$ bits, the maximum for six outcomes.

Binary entropy

For $X \sim \text{Ber}(p)$, entropy depends only on p :

$$H_b(p) = p \log \frac{1}{p} + (1 - p) \log \frac{1}{1 - p}.$$



Experiment	H_b
Fair coin	1
A 6 on a die	0.65
Rain, Seattle	0.977
Rain, Phoenix	0.438
Rain, Sahara	0.027

Maximum entropy is 1 bit

One bit exactly represents the outcome of a fair coin flip.

Your turn

Which has higher entropy: $X \sim \text{Bin}(2, 1/2)$ or Y uniform on $\{0, 1, 2\}$? Compute both.

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Answer

X : outcomes 0, 1, 2 with probabilities $1/4, 1/2, 1/4$, so $H(X) = 1.5$ bits.

Y : uniform on 3 outcomes, so $H(Y) = \log_2 3 \simeq 1.585$ bits.

Y is higher: for a fixed number of outcomes, uniform maximizes entropy.

Surprising outcome, uninformative variable

A lottery: match 5 of 70 plus 1 of 25. Win probability

$$\frac{1}{\binom{70}{5} \cdot 25} = \frac{1}{302,575,350}.$$

- Winning conveys $\log_2 302,575,350 \simeq 28.2$ bits.
- Losing conveys $\simeq 4.8 \times 10^{-9}$ bits.
- Entropy of the win/lose variable: $H_b(1/302,575,350) \simeq 10^{-7}$ bits.

Surprising is not the same as informative

An outcome can be extremely surprising even when the variable carries almost no information on average.

The entropy of English

A letter drawn at random from English text, over 26 letters A–Z.

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Prediction is compression

A better predictor and a lower entropy estimate are the same thing; language models exploit exactly this predictability.

Why “entropy”

Entropy was introduced by Claude Shannon in “A Mathematical Theory of Communication” (1948), where he also popularized the term **bit** (binary digit).

“My greatest concern was what to call it. I thought of calling it ‘information,’ but the word was overly used, so I decided to call it ‘uncertainty.’ When I discussed it with John von Neumann, he had a better idea. Von Neumann told me, ‘You should call it entropy, for two reasons. In the first place your uncertainty function has been used in statistical mechanics under that name, so it already has a name. In the second place, and more important, no one really knows what entropy really is, so in a debate you will always have the advantage.’”

Claude Shannon, as told to Tribus and McIrvine, *Scientific American* (1971), vol. 225, p. 180.

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Coding the alphabet

English text, transmitted one letter at a time over A–Z.

- Fixed length: $\lceil \log_2 26 \rceil = 5$ bits per letter; “TEA” = 15 bits.

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- Fixed length: $\lceil \log_2 26 \rceil = 5$ bits per letter; “TEA” = 15 bits.
- Variable length (Huffman): short codewords for frequent letters, long for rare ones. “TEA” = 000 100 1110 = 10 bits; 4.21 bits/letter on average, versus entropy 4.18.

Letter	Codeword	Length	Freq (%)
E	100	3	12.7
T	000	3	9.1
A	1110	4	8.2
O	1101	4	7.5
N	1010	4	6.7
Q	0011110110	10	0.1
Z	0011110111	10	0.1

Samuel Morse and Alfred Vail arrived at the same principle in 1838: E is a single dot.

Source coding theorem

The ideal code length for an outcome with probability p is $\log_2(1/p)$ bits, exactly the self-information.

Shannon's source coding theorem

Assign roughly $\log_2(1/p)$ bits to an outcome with probability p : short codewords for likely outcomes, long ones for unlikely outcomes. The minimum achievable average code length is the entropy $H(X)$. No lossless code averages fewer than $H(X)$ bits per symbol, whatever the computation.

Averaging the ideal lengths $\log_2(1/p(x))$ over x gives exactly $H(X)$.

Your turn

A source emits A, B, C, D with the probabilities and codewords below. Compute $H(X)$, and check the code.

Symbol	A	B	C	D
Probability	$1/2$	$1/4$	$1/8$	$1/8$
Codeword	0	10	110	111

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Answer

$$H(X) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = 1.75 \text{ bits.}$$

Average code length: $\frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = 1.75$, matching $H(X)$. Each codeword length equals $\log_2(1/p)$ for its symbol.

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Cross-entropy

A code optimized for q assigns about $\log(1/q(x))$ bits to outcome x . If the data actually follow p , the average code length is the **cross-entropy**

$$H(p, q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{q(x)}.$$

DNA example

$p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ over A, C, G, T. Not knowing this, a code for the uniform q spends $\log_2 4 = 2$ bits per base. The best possible is $H(X) = 1.75$ bits.

The penalty for using the wrong distribution:

$$D(p \parallel q) = H(p, q) - H(p) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)},$$

the bits wasted per symbol by coding with q instead of p .

DNA, continued

$D(p \parallel q) = 2 - 1.75 = 0.25$ bits. Term by term: A contributes +0.5, C contributes 0, G and T each contribute -0.125 ; the total is 0.25 bits wasted.

Three properties of relative entropy

- **Non-negative:** no code beats the entropy bound, so $D(p \parallel q) \geq 0$, with equality iff $q = p$.
- **Can be infinite:** if $q(x) = 0$ but $p(x) > 0$, the code for q has no codeword for x . DNA with $q(T) = 0$: $D(p \parallel q) = \infty$.
- **Asymmetric:** $D(p \parallel q) \neq D(q \parallel p)$ in general.

Fair coin versus a biased coin

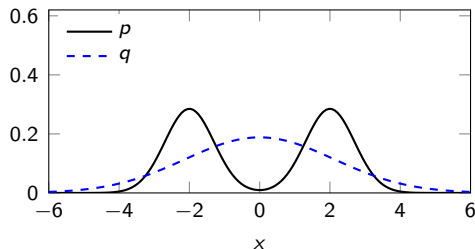
p fair, q with $\Pr(\text{heads}) = 0.9$: $D(p \parallel q) \simeq 0.74$ bits, $D(q \parallel p) \simeq 0.53$ bits.

Relative entropy as a cost function

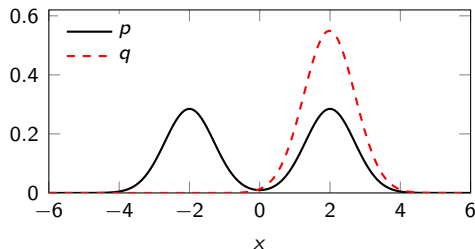
Fitting a model q to data from p :

- Minimizing $D(p \parallel q)$ makes q cover everything p covers (maximum likelihood training of generative models).
- Minimizing $D(q \parallel p)$ lets q sit on one mode and ignore others (variational inference).

minimizing $D(p \parallel q)$



minimizing $D(q \parallel p)$



- For a fixed p , minimizing $D(p \parallel q)$ is the same as minimizing the cross-entropy $H(p, q)$, and this is the form usually used as a loss function.

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Alice, Bob, and traffic: the model

Alice and Bob commute through the same traffic.

- T_O : normal traffic. T_H : heavy traffic.
- A_O, B_O : Alice, Bob on time. A_L, B_L : Alice, Bob late.

Traffic T_t	$\Pr(T_t)$	$\Pr(A_O \mid T_t)$	$\Pr(B_O \mid T_t)$
Normal T_O	0.6	0.95	0.90
Heavy T_H	0.4	0.45	0.15

Late probabilities are the complements, e.g. $\Pr(A_L \mid T_O) = 0.05$.

Independent given traffic: $A \perp\!\!\!\perp B \mid T$

$$\Pr(T_t, A_a, B_b) = \Pr(T_t) \Pr(A_a \mid T_t) \Pr(B_b \mid T_t).$$

The joint distribution of the two arrivals

Average over the two traffic conditions:

$$\Pr(A_a, B_b) = \sum_{t=0}^1 \Pr(T_t) \Pr(A_a \mid T_t) \Pr(B_b \mid T_t).$$

For example, both arrive on time with probability

$$\Pr(A_O, B_O) = 0.6(0.95)(0.90) + 0.4(0.45)(0.15) = 0.54.$$

	Bob on time B_O	Bob late B_L	Total
Alice on time A_O	0.54	0.21	0.75
Alice late A_L	0.06	0.19	0.25
Total	0.60	0.40	1

The arrivals are dependent: $\Pr(A_O, B_O) = 0.54 \neq 0.75(0.60) = 0.45$.

Two arrivals together: joint entropy

How much uncertainty is there in the pair (A, B) ? Write $p(a, b) = \Pr(A_a, B_b)$.

Joint entropy

$$H(A, B) = - \sum_{a,b} p(a, b) \log_2 p(a, b) = \mathbb{E} \left[\log_2 \frac{1}{p(A, B)} \right].$$

Treat the four arrival pairs as the outcomes of one random variable:

$$\begin{aligned} H(A, B) &= 0.54 \log_2 \frac{1}{0.54} + 0.21 \log_2 \frac{1}{0.21} \\ &\quad + 0.06 \log_2 \frac{1}{0.06} + 0.19 \log_2 \frac{1}{0.19} \\ &\simeq 1.652 \text{ bits.} \end{aligned}$$

Separately, $H(A) = H_b(0.75) \simeq 0.811$ and $H(B) = H_b(0.60) \simeq 0.971$ bits.

Adding the entropies counts shared information twice

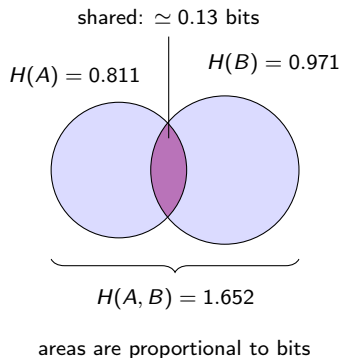
The separate entropies add to

$$H(A) + H(B) \simeq 1.782 \text{ bits.}$$

The joint entropy is

$$H(A, B) \simeq 1.652 \text{ bits.}$$

Their difference is the shared information: approximately 0.13 bits. The joint entropy counts it once.



Observing the pair gives at least as much information as either arrival alone.

Conditional entropy: uncertainty after observing Bob

Bob's arrival changes the distribution of Alice's arrival.

Bob's arrival	Probability	$\Pr(A_O B_b)$	$H(A B_b)$
On time B_O	0.60	$0.54/0.60 = 0.90$	0.469 bits
Late B_L	0.40	$0.21/0.40 = 0.525$	0.998 bits

Conditional entropy averages over the observation

$$H(A | B) = \sum_b \Pr(B_b) H(A | B_b) \simeq 0.60(0.469) + 0.40(0.998) = 0.681 \text{ bits.}$$

Here $H(A | B_b) = H_b(\Pr(A_O | B_b))$.

Before observing Bob, Alice's entropy is 0.811 bits. Afterward it is 0.681 bits *on average*.

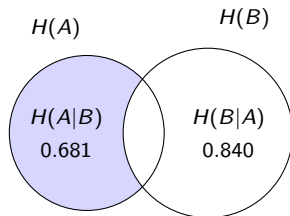
The chain rule for entropy

Describe Bob's arrival first, then Alice's arrival given Bob's.

Joint entropy splits into two parts

$$H(X, Y) = H(Y) + H(X | Y).$$

$$\begin{aligned} H(A, B) &= H(B) + H(A | B) \\ &\simeq 0.971 + 0.681 = 1.652. \end{aligned}$$



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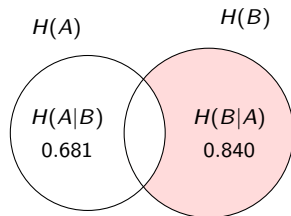
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The other order gives

$$H(A, B) \simeq 0.811 + 0.840.$$

The conditional entropies differ, but the total is the same. All values are in bits.



The overlap: mutual information

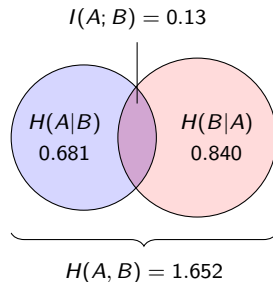
Observing Bob reduces our uncertainty about Alice by about 0.13 bits on average.

Mutual information is symmetric

$$\begin{aligned} I(A; B) &= H(A) - H(A | B) \\ &= H(B) - H(B | A) \\ &= H(A) + H(B) - H(A, B). \end{aligned}$$

$$I(A; B) \simeq 0.811 - 0.681 = 0.13 \text{ bits.}$$

Bob removes the same number of bits about Alice as Alice removes about Bob.



Mutual information measures dependence

If X and Y were independent, their joint distribution would be $p_X(x)p_Y(y)$.

Compare the actual joint distribution with independence

$$I(X; Y) = D(p_{X,Y} \parallel p_X p_Y) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p_X(x)p_Y(y)}.$$

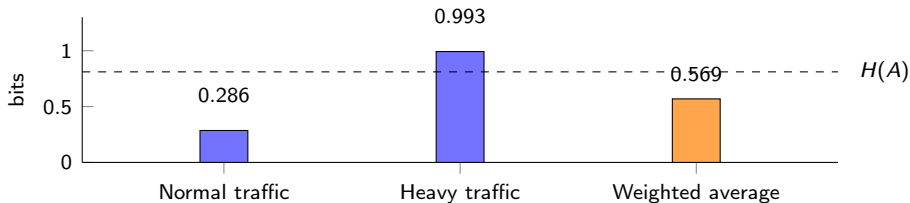
For the arrivals, the actual probability of both being on time is 0.54. Independence would give $0.75(0.60) = 0.45$.

- $I(X; Y) \geq 0$, with equality exactly when $X \perp\!\!\!\perp Y$.
- Thus $H(X | Y) \leq H(X)$: conditioning reduces entropy on average.
- For discrete variables, $I(X; Y) \leq \min\{H(X), H(Y)\}$.

The same averaging principle gives $H(X | Y, Z) \leq H(X | Z)$.

Conditioning helps on average, not in every case

For Alice, $H(A | T_O) = H_b(0.95)$ and $H(A | T_H) = H_b(0.45)$.



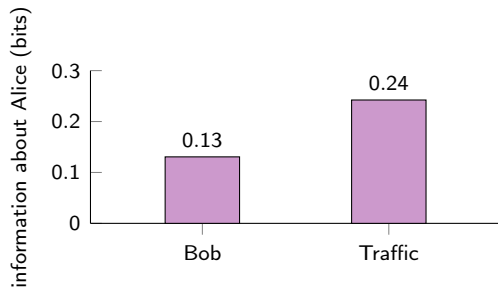
$$H(A | T) = 0.6H_b(0.95) + 0.4H_b(0.45) \simeq 0.569 \text{ bits.}$$

A particular observation can increase uncertainty

Alice's arrival is more uncertain under heavy traffic: $0.993 > 0.811$ bits. The weighted average still satisfies $H(A | T) \leq H(A)$.

Traffic tells us more about Alice than Bob does

Compare the average reduction in uncertainty about the *same* arrival, Alice's.



$$I(A; B) \simeq 0.811 - 0.681 = 0.13,$$

$$I(A; T) \simeq 0.811 - 0.569 = 0.242.$$

Traffic tells us even more about Bob:

$$H(B | T) = 0.6H_b(0.90) + 0.4H_b(0.15)$$

$$\simeq 0.525, \quad I(B; T) \simeq 0.446 \text{ bits.}$$

Alice's on-time probability changes from 0.95 to 0.45 across traffic conditions. Bob's changes from 0.90 to 0.15.

Your turn: rain and a wet sidewalk

X indicates overnight rain. Y indicates a wet sidewalk in the morning. A sprinkler can wet the sidewalk too.

$p(x, y)$	Wet	Dry	$p_X(x)$
Rain	1/4	0	1/4
No rain	1/4	1/2	3/4
$p_Y(y)$	1/2	1/2	1

- 1 Find $H(X | Y)$ by averaging over wet and dry observations.
- 2 Check your answer using $H(X | Y) = H(X, Y) - H(Y)$.
- 3 Find $I(X; Y)$. How much uncertainty about rain does the sidewalk remove?

Wet sidewalk: two ways to find conditional entropy

Average over observations

Dry: rain has probability 0, so the entropy is 0 bits.

Wet: rain has probability $(1/4)/(1/2) = 1/2$, so the entropy is 1 bit.

$$H(X | Y) = \frac{1}{2}(0) + \frac{1}{2}(1) = 0.5.$$

Use the chain rule

The joint probabilities are $1/4, 0, 1/4, 1/2$, so $H(X, Y) = 1.5$ bits.

Wet and dry are equally likely, so $H(Y) = 1$ bit.

$$H(X | Y) = 1.5 - 1 = 0.5.$$

Mutual information

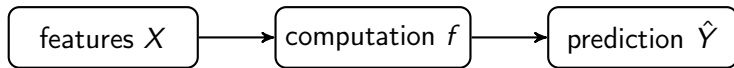
$I(X; Y) = H_b(1/4) - 0.5 \simeq 0.311$ bits. The sidewalk removes about 38% of the initial uncertainty about rain, on average.

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Processing cannot add information about the label

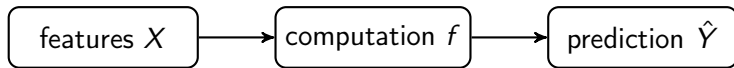
Let X be the features and Y the label. Fix a predictor f , with output $\hat{Y} = f(X)$.



Which is larger, $I(\hat{Y}; Y)$ or $I(X; Y)$?

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Let X be the features and Y the label. Fix a predictor f , with output $\hat{Y} = f(X)$.



Which is larger, $I(\hat{Y}; Y)$ or $I(X; Y)$?

Data processing for a function

$$I(\hat{Y}; Y) \leq I(X; Y).$$

The output cannot contain more information about the label than the input does.

Processing cannot create relevant information.

Why a function cannot add relevant information

For any function g , knowing V also means knowing $g(V)$.

$$\begin{aligned} H(U \mid V) &= H(U \mid V, g(V)) \\ &\leq H(U \mid g(V)). \end{aligned}$$

The inequality follows because conditioning on more variables cannot increase average uncertainty.

Subtract from the same initial uncertainty $H(U)$

$$\underbrace{H(U) - H(U \mid g(V))}_{I(U;g(V))} \leq \underbrace{H(U) - H(U \mid V)}_{I(U;V)}.$$

A summary may discard distinctions in V . It can preserve all the relevant information, or lose some of it.

Markov chains: processing with randomness

A randomized output Z may depend on V without being a function of V alone.

The condition for a Markov chain $U \rightarrow V \rightarrow Z$

$$U \perp\!\!\!\perp Z \mid V, \quad p(z \mid u, v) = p(z \mid v)$$

for conditioning events of positive probability. Given V , the output Z provides no further information about U .

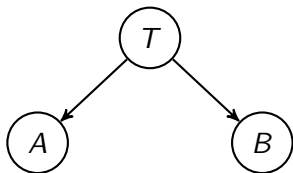
U	V	Z
Original message	What player 2 hears	What player 3 hears
Scene	Photo	Edited photo

In telephone, player 3 hears only player 2. In image editing, the editor uses only the photo and randomness independent of the scene.

Markov order and causal direction

In our arrival model, traffic is the common cause and $A \perp\!\!\!\perp B \mid T$.

The same conditional independence gives both Markov chains:



$$A \rightarrow T \rightarrow B, \quad B \rightarrow T \rightarrow A.$$

The middle variable separates the two ends. These chains do not claim that arrivals cause traffic.

The illness example in the notes

In a hypothetical model, let illness state V generate temperature U and blood oxygen Z *independently given* V . Then $U \rightarrow V \rightarrow Z$ holds. Conditional independence is a modeling assumption, not a consequence of calling V the “full state.”

The data processing inequality

Information about U cannot increase along the chain

If $U \rightarrow V \rightarrow Z$, then $I(U; Z) \leq I(U; V)$.

The same entropy argument applies:

$$H(U | Z) \geq H(U | V, Z) = H(U | V).$$

The equality uses $U \perp\!\!\!\perp Z | V$.

Alice, traffic, and Bob

$$A \rightarrow T \rightarrow B \implies I(A; B) \simeq 0.13 \leq I(A; T) \simeq 0.242 \text{ bits.}$$

Bob's arrival retains about 54% of the information that traffic provides about Alice.

Reversing the chain also gives $I(A; B) \leq I(B; T) \simeq 0.446$ bits.

Your turn: which Markov chains follow?

For each proposed $U \rightarrow V \rightarrow Z$, ask: does $U \perp\!\!\!\perp Z \mid V$ follow?

- a Game of telephone: message, what player 2 hears, what player 3 hears. Each player hears only the preceding player.
- b Scene, photo, edited photo. The edit uses only the photo and independent randomness.
- c Temperature, sick/not sick, blood oxygen. Knowing sick/not sick leaves severity unknown.
- d Temperature, illness state, blood oxygen. Assume independent readings given illness state.
- e Alice's arrival, Bob's arrival, traffic, under our specified model.

Markov chains: answers and assumptions

Chain?	Reason
(a) Yes	The third player's information comes only through the second.
(b) Yes	Given the photo, the output uses no further information about the scene.
(c) Not guaranteed	Unobserved severity may keep the readings dependent within the sick group. A joint model is needed.
(d) Yes	Conditional independence is an explicit assumption. The reverse order also works.
(e) No	In our model, $\Pr(A_O B_O) = 0.90$, but $\Pr(A_O B_O, T_O) = 0.95$. Traffic still changes Alice's distribution.

A story suggests a model. The conditional independence determines whether the chain holds.

Summarizing the data cannot add information about a parameter

Take the Bayesian view: the parameter Θ has a prior, and given $\Theta = \theta$ the sample $X = (X_1, \dots, X_n)$ has distribution $p(x | \theta)$. A **statistic** $S(X)$ is any function of the sample: the sample mean, the sample variance, the largest observation.

The summary is computed from the data

$$\Theta \rightarrow X \rightarrow S(X) \quad \implies \quad I(\Theta; S(X)) \leq I(\Theta; X).$$

Question: usually a summary loses something. Which summaries lose nothing?

Summarizing the data cannot add information about a parameter

Take the Bayesian view: the parameter Θ has a prior, and given $\Theta = \theta$ the sample $X = (X_1, \dots, X_n)$ has distribution $p(x | \theta)$. A **statistic** $S(X)$ is any function of the sample: the sample mean, the sample variance, the largest observation.

The summary is computed from the data

$$\Theta \rightarrow X \rightarrow S(X) \quad \implies \quad I(\Theta; S(X)) \leq I(\Theta; X).$$

Question: usually a summary loses something. Which summaries lose nothing?

Sufficient statistic

$S(X)$ is **sufficient** for Θ if $X \perp\!\!\!\perp \Theta \mid S(X)$ for every prior on Θ , equivalently if

$$\Pr(X = x \mid S(X) = s, \Theta = \theta) = \Pr(X = x \mid S(X) = s).$$

Once the summary is known, the parameter has no further effect on the sample. This is the form we check.

A sufficient statistic for the coin bias

Conditional on $\Theta = \theta$, the n tosses are independent $\text{Ber}(\theta)$ variables. Let $S = \sum_i X_i$ count heads.

$$\text{H T T H T H T T} \quad n = 8, S = 3$$

For any sequence x with k heads,

$$\Pr(X = x \mid \Theta = \theta) = \theta^k (1 - \theta)^{n-k},$$

$$\Pr(S = k \mid \Theta = \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}.$$

Conditioning on the count removes the parameter

$$\Pr(X = x \mid S = k, \Theta = \theta) = \frac{1}{\binom{n}{k}}.$$

For $n = 8, k = 3$, all 56 arrangements are equally likely, for every θ with $\Pr(S = 3 \mid \Theta = \theta) > 0$.

Sufficiency: a summary that preserves relevant information

The count is sufficient, and sufficiency supplies a second Markov chain, running the other way.

The summary is a function of the data

$$\Theta \rightarrow X \rightarrow S(X)$$
$$I(\Theta; S(X)) \leq I(\Theta; X).$$

Sufficiency gives the reverse chain

$$\Theta \rightarrow S(X) \rightarrow X$$
$$I(\Theta; X) \leq I(\Theta; S(X)).$$

The two inequalities force equality

$I(\Theta; S(X)) = I(\Theta; X)$: the summary preserves all information about the parameter.

For the coin, keep the count S and the known sample size n . The maximum-likelihood estimate is S/n .

Outline

- 1 Quantifying uncertainty: Entropy
- 2 Entropy in action: data compression
- 3 The wrong distribution: relative entropy
- 4 More than one variable: Joint entropy and mutual information
- 5 Can processing add relevant information?
- 6 Limits on prediction

Fano's inequality: uncertainty forces prediction errors

Let Y take $m \geq 2$ possible labels. A predictor $\hat{Y} = f(X)$ uses the features X . Its error probability is $P_e = \Pr(\hat{Y} \neq Y)$.

Fano's inequality, with base-2 logarithms

$$H(Y | X) \leq H_b(P_e) + P_e \log_2(m - 1).$$

If we know the prediction, describing the true label requires:

- Whether an error occurred, with entropy at most $H_b(P_e)$.
- On an error, which of the other $m - 1$ labels is correct, using at most $\log_2(m - 1)$ bits of entropy.

Small error therefore requires small remaining uncertainty $H(Y | X)$.

Binary prediction: half a bit forces at least 11% error

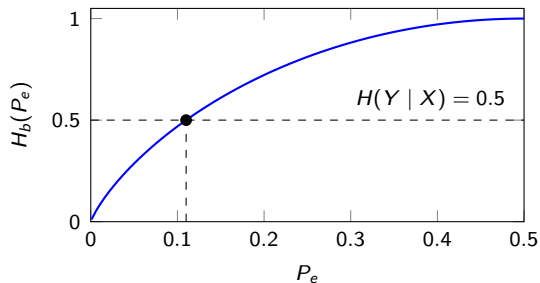
For $m = 2$, Fano gives $H_b(P_e) \geq H(Y | X)$.
If $H(Y | X) = 0.5$ bits, solve

$$H_b(t) = 0.5, \quad 0 \leq t \leq \frac{1}{2}.$$

The smaller root is $t \simeq 0.1100$.

Every predictor has an error floor

$P_e \geq 0.1100$, so accuracy is at most about 89%.



A simpler bound for many labels

Since $H_b(P_e) \leq 1$ and $\log_2(m-1) \leq \log_2 m$,

$$H(Y | X) \leq 1 + P_e \log_2 m.$$

A weaker, explicit error bound

$$P_e \geq \frac{H(Y | X) - 1}{\log_2 m} = \frac{H(Y) - I(X; Y) - 1}{\log_2 m}.$$

The features provide $I(X; Y)$ bits about the label. Fano turns the remaining uncertainty into a lower bound on prediction error.

When $H(Y | X) \leq 1$, this weaker bound gives no positive error floor. Use the original inequality, as in the binary example.

Next-character prediction: the 27-symbol model

Let Y be the next character and X the preceding text. Use $m = 27$ symbols: the 26 letters and a space.

Take $H(Y | X) \simeq 1$ bit as the entropy estimate

$$H_b(P_e) + P_e \log_2 26 \geq 1.$$

The smallest admissible error is $P_e \simeq 0.1078$, giving an accuracy upper bound of about 89.2%.

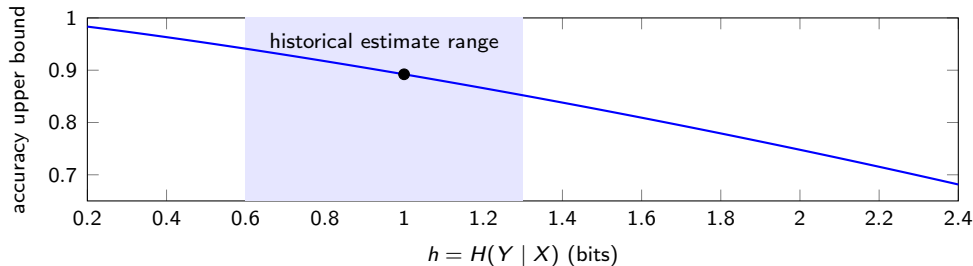
No predictor, however large the model and however much text it was trained on, can guess the next character correctly more than about 89% of the time.

The ceiling depends on the entropy estimate alone. A better language model leaves it unchanged, and a better estimate of the entropy of English moves it.

Historical entropy estimate: Shannon, Prediction and Entropy of Printed English (1951).

How the entropy estimate changes the accuracy bound

For $m = 27$, invert $H_b(t) + t \log_2 26 = h$ on $0 \leq t \leq 26/27$.



Shannon estimated between 0.6 and 1.3 bits per letter for a context of 100 letters, putting the ceiling on accuracy between 94% and 85%.

Shannon (1951): long-context estimate range. A separate first-guess experiment scored $89/129 \simeq 69\%$, with the correct character revealed after each guess.

Information about a fixed parameter

In frequentist inference, θ^* is fixed and the sample $X \sim p(x; \theta^*)$ is random.

A random unknown

A prior makes Θ random.

$$I(\Theta; X)$$

measures information about Θ under that joint distribution.

A fixed unknown

Treating θ^* as a constant gives zero mutual information with X .

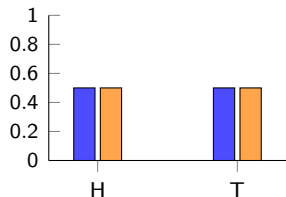
Instead, compare $p(x; \theta)$ with distributions at nearby parameter values.

The question for estimation

How much does the distribution of the data change when the parameter changes?

Distinguishing two coin biases

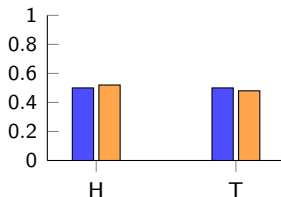
Compare one toss from $\text{Ber}(0.50)$ with one from $\text{Ber}(\theta')$. Blue: $\theta = 0.50$. Orange: the labeled θ' . KL uses natural logarithms (nats).



$$\theta' = 0.50$$

$$D = 0$$

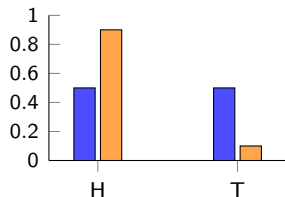
Same distribution.



$$\theta' = 0.52$$

$$D \simeq 0.0008$$

Nearby biases are harder to distinguish.



$$\theta' = 0.90$$

$$D \simeq 0.5108$$

The distributions differ much more.

If two *different* parameter values give the same distribution, data cannot distinguish them. A Bernoulli bias is identifiable: equal distributions imply equal biases.

Fisher information: local curvature of relative entropy

For a regular scalar-parameter model, expand the divergence near θ using natural logarithms:

The leading local change is quadratic

$$D(p_\theta \parallel p_{\theta+\delta}) = \frac{1}{2}\mathcal{I}(\theta)\delta^2 + o(\delta^2).$$

Equivalently,

$$\mathcal{I}(\theta) = \lim_{\delta \rightarrow 0} \frac{2}{\delta^2} D(p_\theta \parallel p_{\theta+\delta}).$$

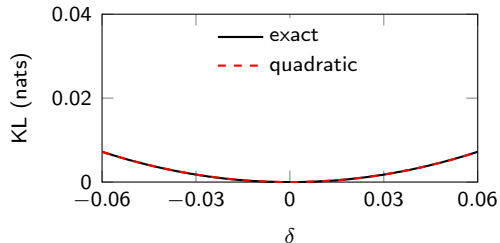
Larger curvature means that the same small change in θ produces a larger change in the data distribution.

This expansion requires regularity. Parameter-dependent support, as in $\text{Uni}[0, \theta]$, can invalidate it.

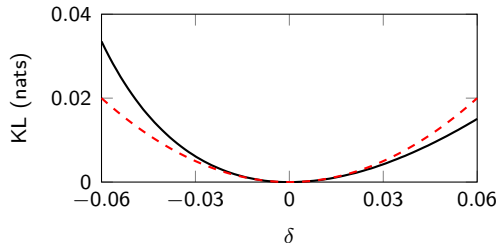
Fisher information for a Bernoulli observation

For $0 < \theta < 1$, the local expansion gives $\mathcal{I}(\theta) = 1/[\theta(1 - \theta)]$.

$$\theta = 0.5, \mathcal{I}(\theta) = 4$$



$$\theta = 0.1, \mathcal{I}(\theta) \simeq 11.11$$



The panels use identical scales. Near $\theta = 0.1$, the divergence has greater curvature than near $\theta = 0.5$. The quadratic approximation describes the immediate neighborhood.

Fisher information will give estimation bounds in the parameter-estimation chapter, under suitable assumptions on the model and estimator.

Summary: uncertainty and compression

- $H(X)$ measures average uncertainty and is the minimum achievable average code length, by the source coding theorem.
- $H(p, q) = H(p) + D(p \parallel q)$ separates intrinsic uncertainty from the penalty for using the wrong distribution.
- $H(X, Y) = H(Y) + H(X \mid Y)$ describes a pair one variable at a time.
- $I(X; Y) = H(X) - H(X \mid Y)$ is the average reduction in uncertainty from observing the other variable.

Summary: processing, prediction, and estimation

- Data processing: $U \rightarrow V \rightarrow Z$ implies $I(U; Z) \leq I(U; V)$.
- Sufficiency: a statistic preserves all information about the parameter when $X \perp\!\!\!\perp \Theta \mid S(X)$ for every prior.
- Fano: remaining label uncertainty gives a lower bound on prediction error.
- Fisher information: local KL curvature measures sensitivity to a fixed parameter and supports estimation bounds under regularity conditions.

Next: frequentist inference and maximum likelihood, Chapter 2 (Sep 8).

For more: the Short Lecture on Information Theory, or Mathematics of Information.